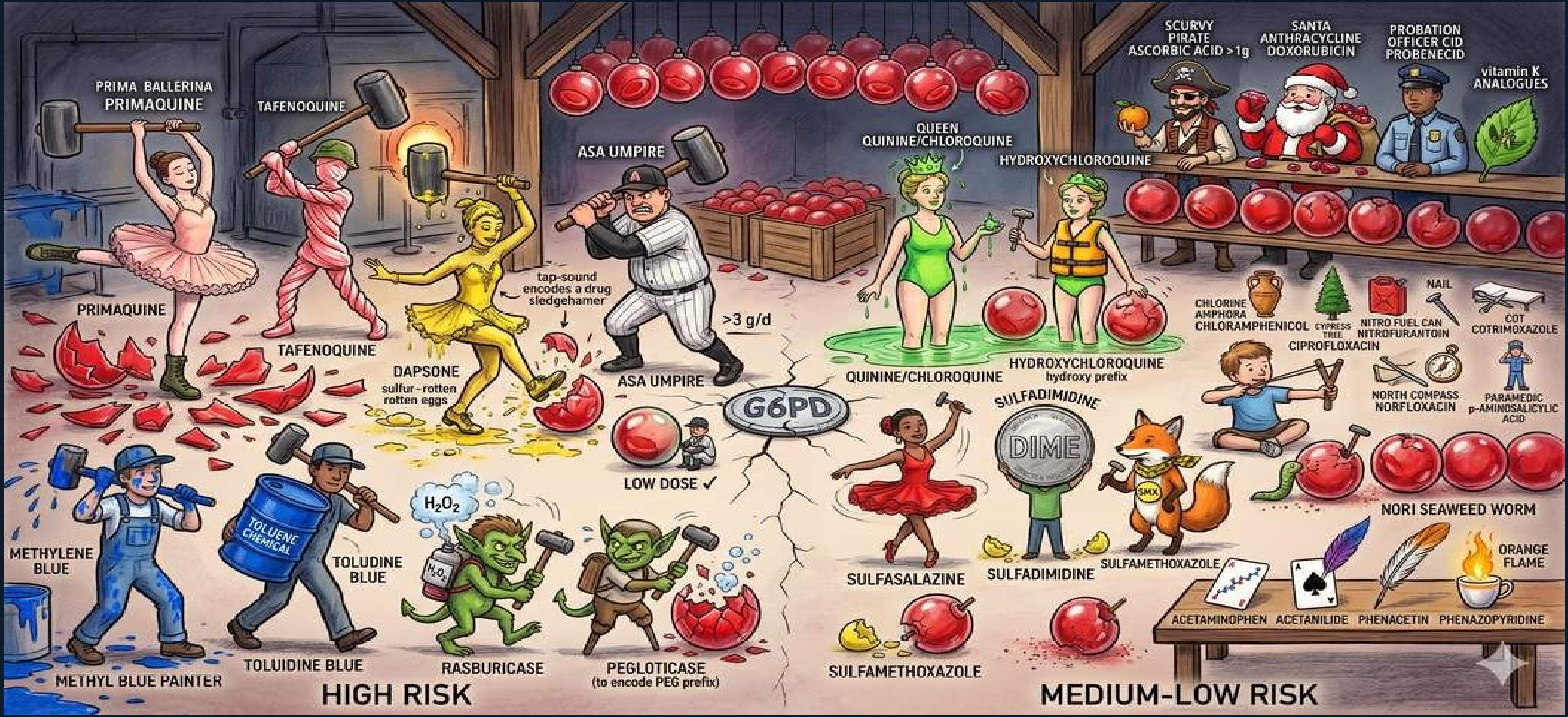
G6PD DRUG RISK ROOM — High Risk vs Medium-Low Risk Drugs



Chapter: G6PD — Drug Classification, Oxidative Triggers, Clinical Risk Stratification

G6PD Drugs

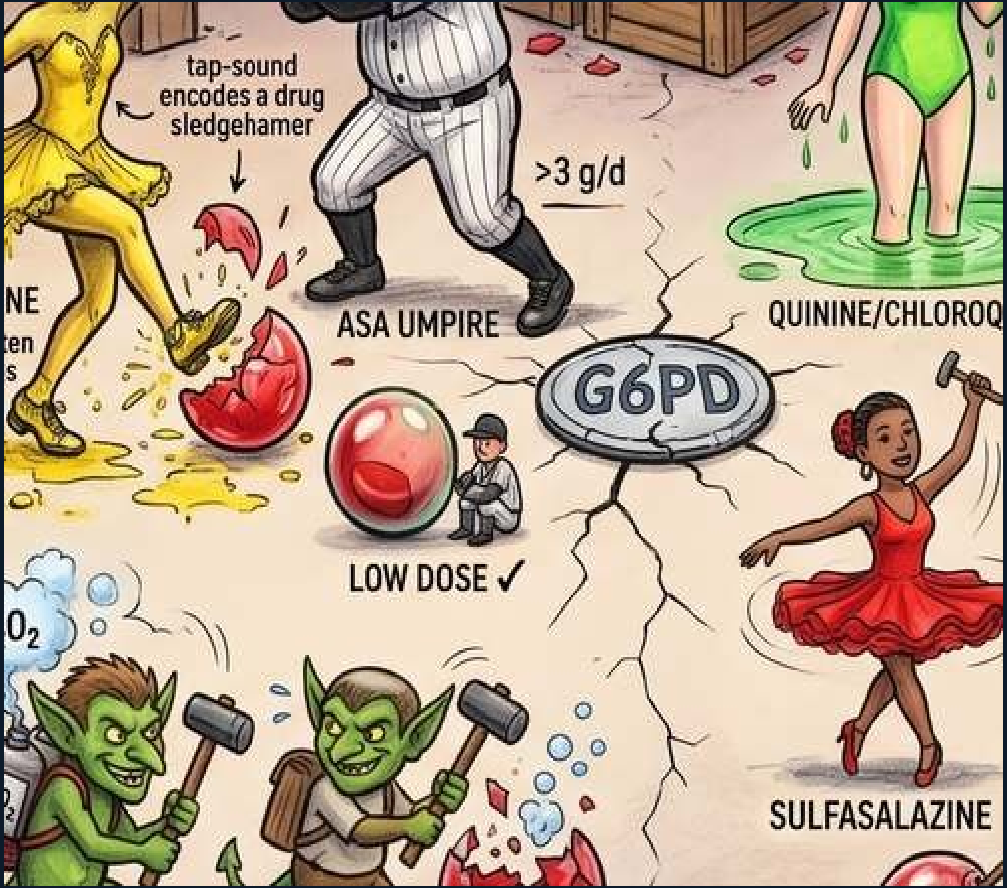
High Risk Zone — Left Side

HIGH RISK drugs in G6PD deficiency — avoid in all variants:

PRIMAQUINE (Prima Ballerina): elegant and deadly in G6PD deficiency. Classic G6PD trigger. 8-aminoquinoline class. Used for malaria radical cure (P. vivax). Must test G6PD before prescribing.

TAFENOQUINE: newer 8-aminoquinoline with even longer half-life. MUST TEST G6PD FIRST — single dose with very long effect window. Fatal cases reported in severe G6PD deficiency.

DAPSONE (Yellow sulfur-rotten eggs figure): sulfone antibiotic. Used for Pneumocystis prophylaxis, dermatitis herpetiformis. H_2O_2 generation in G6PD-deficient RBCs.



G6PD Drugs

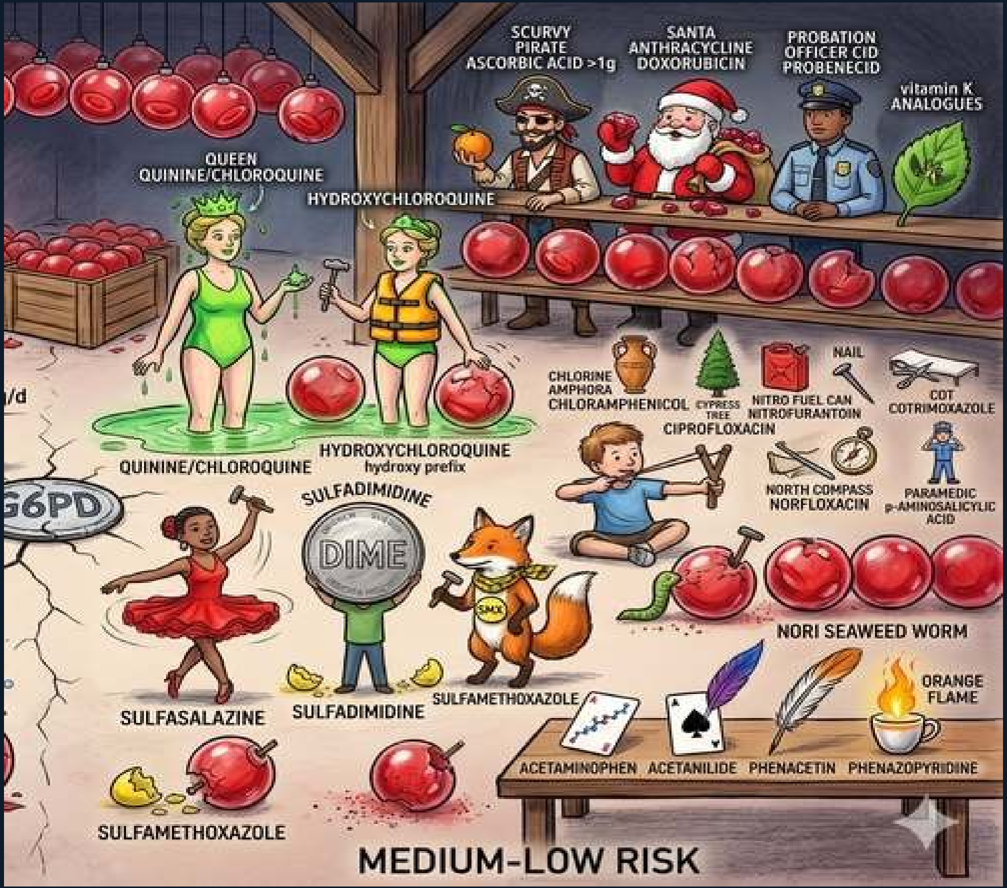
The G6PD Coin — Safe Threshold

The G6PD COIN in the center represents the residual enzyme activity threshold.

The center zone shows where LOW DOSE aspirin is safe (✓) — the coin divides what is safe from what is risky.

METHYLENE BLUE PAINTER: paradoxically, methylene blue (used to treat methemoglobinemia) REQUIRES G6PD to work — it oxidizes Hb but the reduction back to normal methemoglobin requires NADPH from G6PD. In G6PD-deficient patients, methylene blue CANNOT BE USED for methemoglobin treatment → use ascorbic acid instead.

TOLUIDINE BLUE: industrial chemical, similar mechanism to methylene blue.



Medium-Low Risk Zone

MEDIUM-LOW RISK drugs — use with caution, generally safe in mild variants (G6PD A⁻ type) but monitor in severe variants:

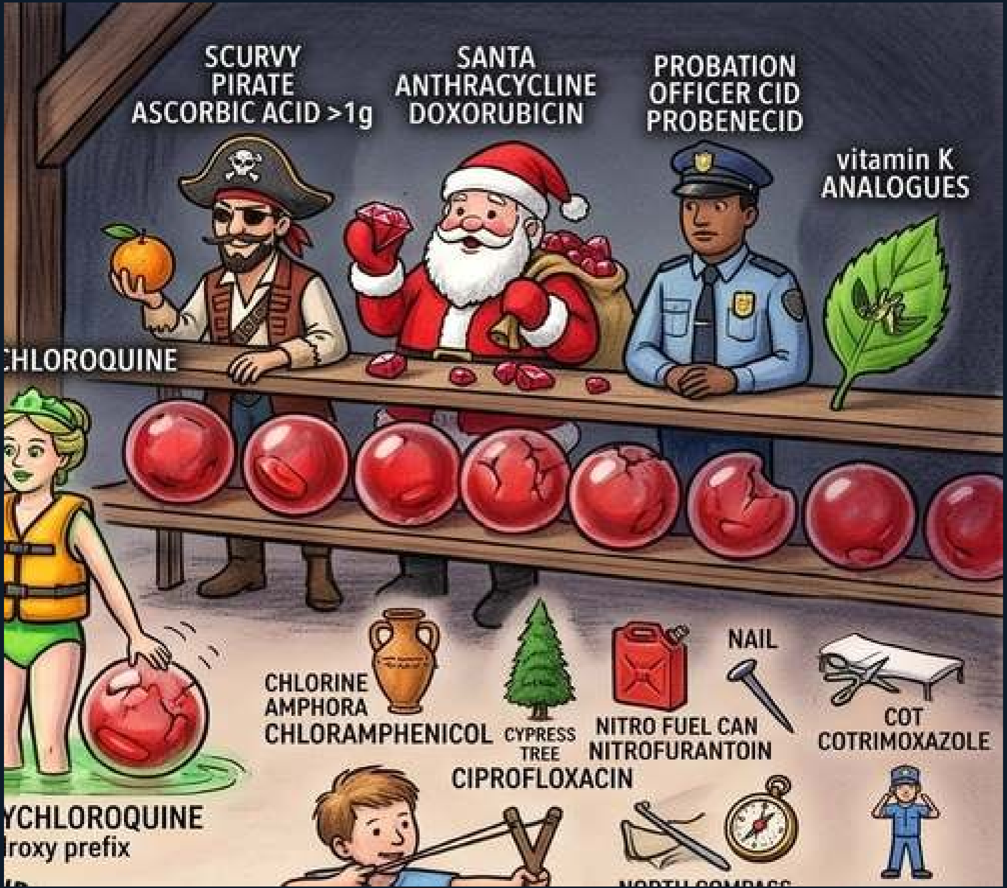
QUININE/CHLOROQUINE (Queen + hydroxyl-life jacket): medium risk. Quinine for malaria. Chloroquine for malaria + lupus/rheumatoid.

HYDROXYCHLOROQUINE: safer than chloroquine. The "hydroxy prefix" = slightly less oxidative.

SULFASALAZINE (Red ballerina on dime): sulfa component → medium risk in severe G6PD variants.

SULFADIMIDINE (dime): sulfonamide antibiotic.

SULFAMETHOXAZOLE (SMX fox): the sulfamethoxazole



G6PD Drugs

Miscellaneous Risk Drugs & Substances

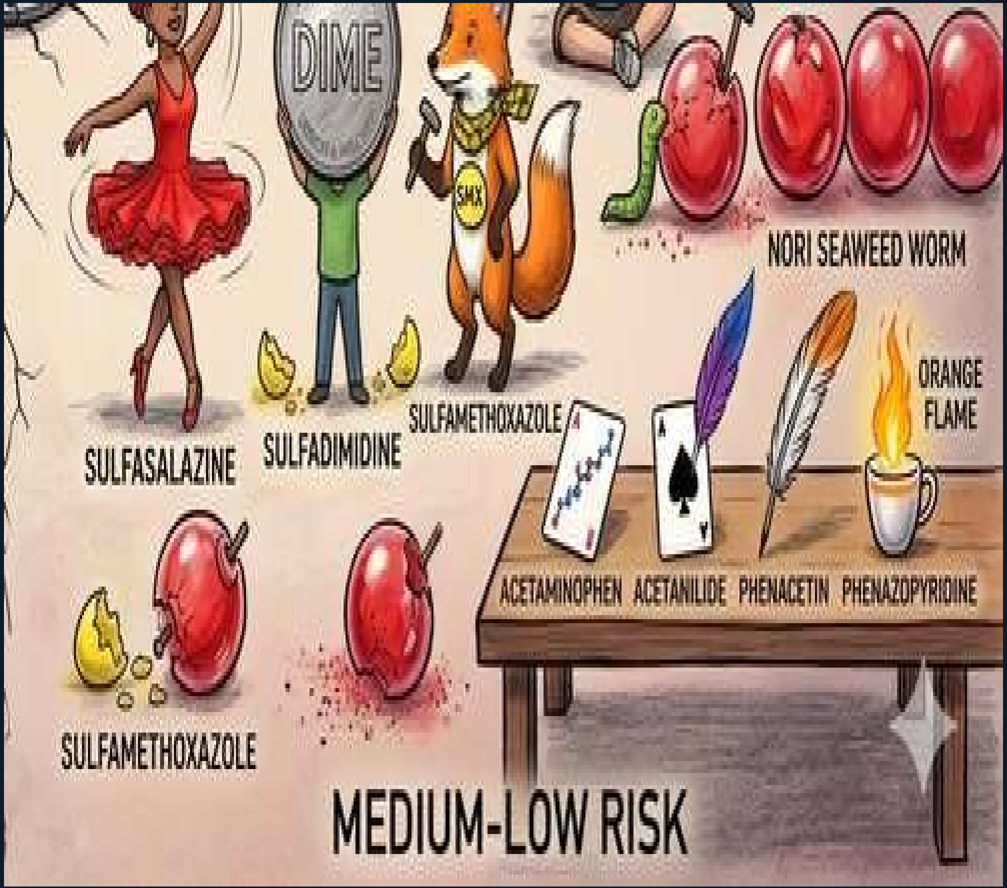
TOP SHELF (dangerous items with skulls/crosses):

SCURVY PIRATE — ASCORBIC ACID >1g: at high doses (>1g IV), ascorbic acid generates free radicals in G6PD-deficient RBCs. Paradoxically the antioxidant becomes pro-oxidant at extreme doses.

ANTHRACYCLINE / DOXORUBICIN (Santa with bag): anthracyclines generate reactive oxygen species → risk in severe G6PD deficiency. Caution in chemotherapy protocols.

PROBENECID (Probation Officer): uricosuric agent. Mild-moderate risk.

VITAMIN K ANALOGUES: synthetic vitamin K (menadione) is oxidative; phytonadione (K1) is safe.



G6PD Drugs

Low Risk Drugs — Bottom Shelf

LOW-RISK / GENERALLY SAFE drugs in G6PD deficiency:

ACETAMINOPHEN / PARACETAMOL: SAFE at standard doses. NOT an oxidative agent. Common misconception that it's contraindicated.

ACETANILIDE: the precursor to acetaminophen. Historically shown to cause methemoglobinemia through the aniline metabolite, but acetaminophen itself is safe.

PHENACETIN: metabolized to acetaminophen. Historical concern but largely replaced.

PHENAZOPYRIDINE (Orange flame): azo dye urinary analgesic. MEDIUM risk — listed here as borderline. Can cause methemoglobinemia + hemolysis at high

G6PD DRUGS — BOARD SUMMARY

Avoid Always

Primaquine, tafenoquine, dapsone, rasburicase, pegloticase, methylene blue.

ASA Rule

Low-dose aspirin (≤100mg) = SAFE. High dose (>3g/d) = risky. Know the threshold.

Methylene Blue Trap

Cannot treat methemoglobin with methylene blue in G6PD deficiency. Use IV ascorbic acid.

Medium Risk

Quinine, hydroxychloroquine, sulfonamides (SMX), nitrofurantoin, ciprofloxacin.

Vitamin K

Phytonadione (K1) SAFE. Menadione (K3) DANGEROUS — oxidative.

Acetaminophen

SAFE in G6PD deficiency. Common misconception it is contraindicated.

Severity Matters

A^- (African) = mild, more drugs tolerable. MED (Mediterranean) = severe, avoid all listed.

Screen Then Prescribe

Test G6PD before primaquine or tafenoquine (8-aminoquinolines). Fatal cases if missed.